ACADEMIC PROJECT REPORT · DATA SCIENCE & MACHINE LEARNING

Rakuten France Multimodal Product Classification

Systematic exploration, deep learning progression, Grad-CAM visual interpretability, and out-of-fold late fusion across 84,916 e-commerce listings.

Key Results & Findings

- **Strict Out-of-Fold Protocol:** A locked 80/20 stratified split (67,932 train, 16,984 validation). Meta-models and soft-voting weights are evaluated with 5-fold cross-validation on the validation set, so no product is scored by a model that learned from it.
- **Multimodal Synergy:** Random Forest late fusion reaches **0.8988 Weighted F1** (Macro F1 0.8856, 1,715 errors), cutting 367 errors (–17.6%) compared to CamemBERT alone and 941 errors (–35.4%) compared to calibrated SVM.
- **Rescuing Missing Descriptions:** On listings lacking descriptions (35.0% of the catalog), fusion gains **+3.90 F1 points over CamemBERT** and **+9.43 F1 points over LinearSVC**, proving that the visual stream decisively compensates for text deficits.
- **Computer Vision Optimization:** From a frozen ResNet18 baseline (0.4818 F1) to ConvNeXt-Tiny with smart white-border cropping (0.6998 F1), **+2.59 points** over the standard ConvNeXt-Tiny.
- **Interpretability:** On the examples shown, Grad-CAM shows ConvNeXt looking at the product rather than the background. Random Forest feature importance relies 66% on CamemBERT, 23% on SVM, and 11% on ConvNeXt.

Project: Rakuten France Product Classification Challenge
Authors: Data Science & Machine Learning Team
Repository: `liora-capstone`

Evaluation Split: Fixed stratified 80/20 split (16,984 validation products)
Core Models: Calibrated LinearSVC, CamemBERT-base, ConvNeXt-Tiny, Late Stacking
Submission Date: November 2026

1. Executive Summary and Problem Formulation

Modern e-commerce platforms process millions of seller-generated product listings daily. Accurate, automated categorization is foundational for catalog organization, search retrieval, facet navigation, and recommendation engines. The **Rakuten France Multimodal Product Classification Challenge** requires categorizing listings into **27 distinct product types** (`prdtypecode`) using noisy, seller-provided text (title and optional description in French) and a single 500 × 500 pixel catalog image.

84,916	27	35.1%	0.8988
PRODUCTS ANALYZED	PRODUCT CATEGORIES	MISSING DESCRIPTIONS	FINAL WEIGHTED F1

Text is the strongest single modality for e-commerce classification. However, text models struggle when seller titles are brief, keyword-stuffed, or linguistically ambiguous. The visual stream provides essential orthogonal information in these failure modes—particularly across book, magazine, toy, and gaming categories.

To establish an empirical floor, the **Zero-Rule Majority Class Baseline** assigns every sample to the largest category (Pool, code 2583). On the 16,984 validation products, this naive baseline achieves an accuracy of **12.02%**, a Weighted F1 of **0.0258**, and a Macro F1 of **0.0080**. Our final multimodal stacking architecture reaches **0.8988 Weighted F1** (Macro F1 0.8856), reducing errors to 1,715 listings.

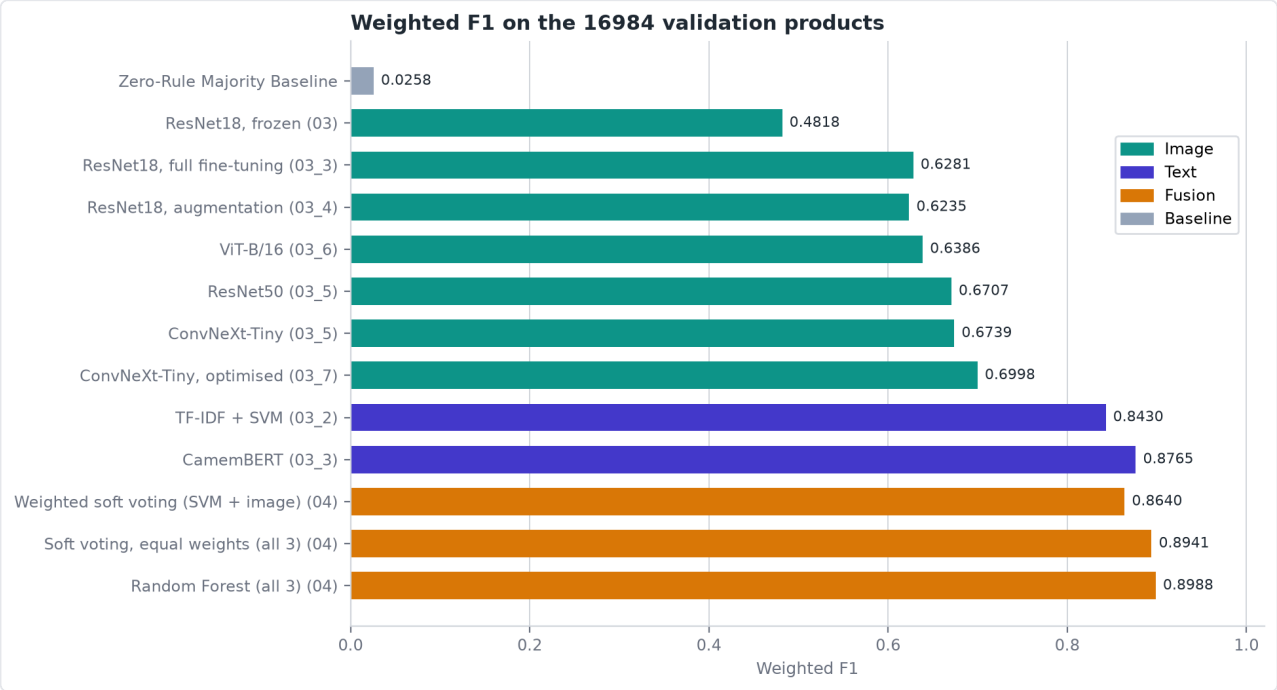


Figure 1: Weighted F1 benchmark progression from the naive majority baseline to deep unimodal models and multimodal late fusion on the locked 16,984 validation products.

2. Exploratory Data Analysis & Data Contract

2.1 Class Distribution and Imbalance

The dataset comprises **84,916 labelled products** across 27 product categories. The distribution exhibits substantial class imbalance:

- The largest category is **Pool (code 2583)** with **10,209 products** (12.02% of the catalog).
- The smallest category is **Wargames (code 1180)** with **764 products** (0.90% of the catalog).

- The imbalance ratio between the largest and smallest category is **13.36 to 1** (~13.4:1). Seven categories contain fewer than 1,000 products.
- Due to this skew, overall accuracy is an inadequate metric. We evaluate models using **Weighted F1** (the official challenge metric) and report **Macro F1** to monitor performance on minority classes.

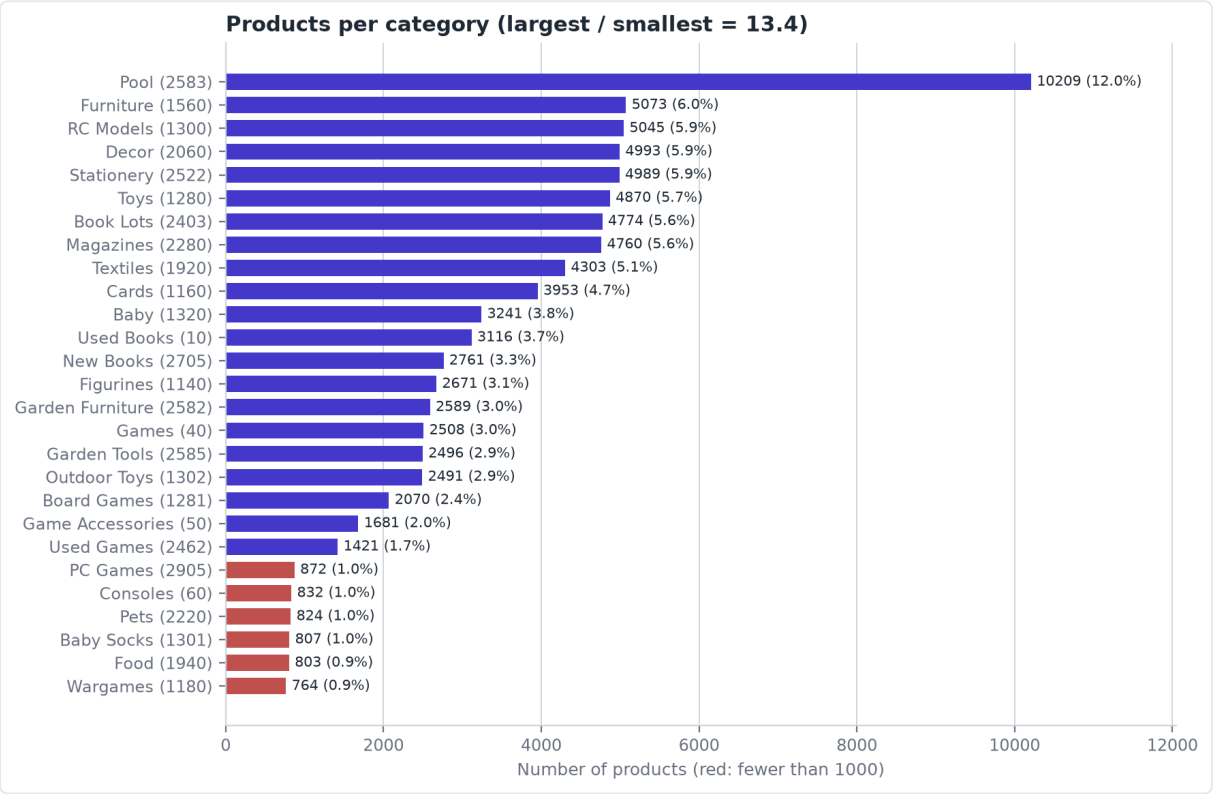


Figure 2: Products per category across the catalog. Red bars indicate the seven minority categories containing fewer than 1,000 listings.

Disclaimer on Category Nomenclature: The challenge dataset identifies classes solely by numeric codes (`prdtypecode`). Category names throughout this report (e.g., *Used Books*, *Wargames*, *Pool*) represent our team's interpretation based on systematic catalog exploration and domain analysis; they are not official Rakuten labels.

Code	Category Name (Team Interpretation)	Full Count	Validation Count	Catalog Share	Missing Desc. %
10	Used Books	3,116	623	3.67%	89.2%
40	Games	2,508	502	2.95%	65.5%
50	Game Accessories	1,681	336	1.98%	28.3%
60	Consoles	832	166	0.98%	16.0%
1140	Figurines	2,671	534	3.15%	64.6%
1160	Cards	3,953	791	4.66%	91.2%
1180	Wargames	764	153	0.90%	80.0%
1280	Toys	4,870	974	5.74%	21.9%
1281	Board Games	2,070	414	2.44%	23.8%
1300	RC Models	5,045	1,009	5.94%	23.3%
1301	Baby Socks	807	161	0.95%	13.0%
1302	Outdoor Toys	2,491	498	2.93%	11.1%

Code	Category Name (Team Interpretation)	Full Count	Validation Count	Catalog Share	Missing Desc. %
1320	Baby	3,241	648	3.82%	34.0%
1560	Furniture	5,073	1,015	5.97%	3.5%
1920	Textiles	4,303	861	5.07%	4.8%
1940	Food	803	161	0.95%	5.6%
2060	Decor	4,993	999	5.88%	5.7%
2220	Pets	824	165	0.97%	8.0%
2280	Magazines	4,760	952	5.61%	93.3%
2403	Book Lots	4,774	955	5.62%	97.4%
2462	Used Games	1,421	284	1.67%	96.3%
2522	Stationery	4,989	998	5.88%	23.8%
2582	Garden Furniture	2,589	518	3.05%	2.4%
2583	**Pool**	**10,209**	**2,042**	**12.02%**	**8.8%**
2585	Garden Tools	2,496	499	2.94%	7.9%
2705	New Books	2,761	552	3.25%	37.1%
2905	PC Games	872	174	1.03%	0.0%

2.2 Text Exploration: Vocabulary, Lengths, and Missingness

- **Title (`designation`):** Available for 100% of products. The median length is 11 words (interquartile range 7–14; mean 11.6 words). Titles are dense with brand names, model numbers, and technical attributes.
- **Description (`description`):** Missing for **29,800 products (35.1% of the catalog)**. In the validation split, 5,936 products (34.95%) lack descriptions. Missingness is highly concentrated in book-related categories: **Book Lots (2403) has 97.4% missing descriptions**, New Books (2705) has 82.5%, and Magazines (2280) has 74.1%. In these categories, unimodal text models are forced to operate purely on brief titles.
- **Formatting Noise:** Raw HTML entities (e.g., `&`, `é`) occur in 41.3% of descriptions, while raw structural tags (`<p>`, `<div>`, `<br>`) appear in 18.4%.
- **Duplicate Titles:** 4,187 products (4.9%) share an identical title string with at least one other catalog entry. In 160 clusters, identical titles map to different product categories (e.g., promotional merchandise vs. media).

2.3 Image Exploration: Dimensions and Border Padding

- **Dimensions & Format:** All 84,916 images are 500 × 500 pixel RGB JPEGs.
- **The White Border Problem:** Many catalog images show a small product in the middle of a large white background. Resizing the whole image to 224 × 224 pixels makes the product very small and loses logos and printed text.
- **Perceptual Duplicates:** Exact SHA-256 and perceptual hash (pHash) analysis reveals duplicate images used across different product listings (e.g., generic placeholder images and multi-pack variants).

2.4 Exact Column Usage Across Model Tiers

To ensure absolute architectural reproducibility, the table below specifies the exact input columns consumed by each model family:

Model Tier	Modality	Raw Input Columns Used	Feature Representation / Transformation
ResNet18 / ConvNeXt-Tiny	IMAGE	`imageid`, `productid`	Loads `image_{imageid}_product_{productid}.jpg`, applies smart white-border crop, resizes to 224 × 224 RGB tensor.
TF-IDF + LinearSVC	TEXT	`designation`, `description`	Cleaned and concatenated into `text_combined`; binary TF-IDF unigram + bigram matrix (460,326 features).
CamemBERT-base	TEXT	`designation`, `description`	Cleaned and concatenated into `text_combined`; tokenized via SentencePiece BPE, truncated to 128 tokens.
Soft Voting (Equal / Weighted)	FUSION	Model output probabilities	Probability vectors from CamemBERT (27), LinearSVC (27), and ConvNeXt-Tiny (27).
Random Forest Stacking	FUSION	**Model output probabilities**	**Concatenated 81-dimensional vector: $[P_{\text{CamemBERT}} \parallel P_{\text{SVM}} \parallel P_{\text{ConvNeXt}}]$.**

3. Preprocessing and Validation Protocol

3.1 Locked Stratified Split

> **Locked 80/20 Validation Split:** A single, globally stratified 80/20 train/validation partition (**67,932 training products, 16,984 validation products**, `random_state=42`) was fixed at project inception (`data_split.csv`). All text, image, and fusion models strictly evaluated on this exact held-out split.

To prevent cross-split contamination, an SHA-256 hash deduplication pass verified that no training image file identical to a validation image was retained in the training set.

3.2 Rigorous Out-of-Fold Protocol for the 16,984 Validation Rows

A critical requirement of this project is absolute transparency regarding how the 16,984 validation products were handled:

1. **Unimodal Training:** The text classifiers (LinearSVC, CamemBERT) were trained only on the 67,932 training rows, and ConvNeXt-Tiny on the 62,312 training images left after removing duplicate and conflicting images. None of them ever saw a single row or image from the 16,984 validation set during training.
2. **Probability Extraction:** The trained unimodal models generated out-of-sample probability distributions for each of the 16,984 validation products.
3. **5-Fold Stratified CV on Validation:** To train the meta-classifiers without data leakage, the 16,984 validation products were divided into **5 stratified folds** (`CV_FOLDS=5`, `random_state=42`):
   - In each fold, the meta-model (Random Forest) was trained on 4/5 of the validation rows (~13,587 products).
   - For weighted soft voting, the weight combinations (α) were chosen by grid search strictly on the 4/5 split.
   - Predictions were then made strictly on the held-out 1/5 fold (~3,397 products).
4. **Final Metric Aggregation:** The reported scores (0.8988 Weighted F1, 1,715 errors) are aggregated strictly from these **out-of-fold predictions** across all 16,984 products. No meta-model or voting rule was ever evaluated on rows it was fitted on.

3.3 Text Preprocessing Pipeline

1. **Entity Decoding:** Multi-pass HTML unescaping converts entities (e.g., `&` → `&`, `é` → `é`).
2. **Tag Stripping:** BeautifulSoup strips HTML markup, substituting spaces to preserve word boundaries.
3. **Whitespace Normalization:** Lowercasing and whitespace collapsing; accents are retained for French linguistic fidelity.

4. **Field Merging:** Designation and description are merged into `text_combined`. Missing descriptions are handled seamlessly without artificial tokens.

3.4 Image Preprocessing: Smart White-Border Cropping

To overcome the white-border pathology identified in Section 2.3, we developed a deterministic **smart white-border cropping algorithm**:

- Each image is turned to greyscale. Pixels darker than a grey level of 245 count as product; lighter pixels count as white background.
- If the non-background bounding box covers less than **75%** of the canvas, the algorithm extracts a square crop centered on the product bounding box at native resolution (at least 224 pixels), with 12 pixels of padding around the product.
- Products that naturally fill the frame are preserved without distortion.
- All cropped images are resized to 224 × 224 RGB tensors and normalized using ImageNet channel statistics.

4. Text Classification Progression

4.1 Classical Text Models and Tuning

Using 3-fold cross-validation on the training set [notebook `03_2_text_optimisation.ipynb`], we evaluated three classical linear classifiers on TF-IDF features:

Classifier	3-Fold CV Weighted F1	Empirical Observations
Complement Naive Bayes	0.7600	Fast, handles class imbalance via complement rule, but lags 6.8 points behind LinearSVC.
Logistic Regression (SGD)	0.8084	Outputs direct probabilities, but lags 2.0 points behind LinearSVC.
LinearSVC	**0.8284**	**Superior maximum-margin separation in high-dimensional sparse text spaces. Selected as primary linear model.**

In notebook `03_2_text_optimisation.ipynb`, we systematically tuned the LinearSVC pipeline:

- **Vocabulary Dimension:** Expanding from the baseline's 50,000 feature limit to the full vocabulary (**460,326 unigrams and bigrams**, `min_df=2`) gained **+1.2 F1 points** on CV. Rare brand names and model numbers reside in the long tail.
- **Binary TF-IDF:** Clamping term frequency to binary presence/absence (`binary=True`) yielded a **+0.5 F1 point gain**. Binary weighting neutralizes keyword stuffing by sellers who repeat terms across descriptions.
- **Regularization:** A grid search over $C \in \{0.1, 0.5, 1.0, 2.0\}$ confirmed $C = 1.0$ as optimal.
- **Platt Scaling Calibration:** Because LinearSVC produces unbounded signed distances rather than probabilities, we wrapped it in `CalibratedClassifierCV(cv=3, method='sigmoid')`. On the 16,984 validation rows, calibrated LinearSVC achieved **0.8430 Weighted F1** (Macro F1 0.8326, 2,656 errors), providing reliable probability vectors for late fusion.

4.2 Transformer Fine-Tuning: CamemBERT

To capture semantic context and French syntactic nuance, we fine-tuned **CamemBERT** (`almanach/camembert-base`, 110 million parameters) [notebook `03_3_text_camembert.ipynb`].

- **Training Setup:** Fine-tuned for 2 epochs on the 67,932 training texts; AdamW optimizer with learning rate 5×10^{-5}, weight decay 0.01, batch size 16, 6% linear warmup followed by linear decay, sequence length 128 tokens.
- **Epoch 1:** Training loss dropped from 2.64 to 0.58; validation loss was 0.501, achieving **0.8527 Weighted F1**.

- **Epoch 2:** Training loss reached 0.34; validation loss declined to 0.427, reaching **0.8765 Weighted F1** (Macro F1 0.8594, 2,082 errors). Continued reduction in validation loss confirmed healthy generalization without overfitting.
- **Comparison to LinearSVC:** CamemBERT reduced errors from 2,656 to 2,082 (**574 fewer errors, a 21.6% reduction**). Major gains occurred across book categories: Used Books (+12.3 F1 points), Magazines (+10.9), New Books (+10.7), and Book Lots (+10.0). LinearSVC outperformed CamemBERT only on Wargames (–8.4 points for CamemBERT), where exact rare unigrams (e.g., *warhammer*, *heroclix*) govern classification.

5. Computer Vision Progression & Grad-CAM Analysis

5.1 Deep Learning Progression Across Backbones

Classifying e-commerce products from images alone is inherently challenging due to packaging similarities and canvas padding. Over multiple systematic experiments [notebooks `03_image_fineTuning/`], we progressed from a frozen ResNet18 baseline to an optimized modern ConvNeXt-Tiny architecture:

Notebook	Architecture	Training Strategy & Pipeline Enhancements	Weighted F1	Macro F1	Errors
03_baseline	ResNet18	Frozen ImageNet backbone; linear classification head only.	0.4818	0.4231	8,462
03_3	ResNet18	Full end-to-end fine-tuning across all convolutional layers.	0.6281	0.5943	6,315
03_4	ResNet18	Augmentation (random crops, flips, rotation, colour jitter, random erasing), label smoothing and weight decay.	0.6235	0.5876	6,377
03_6	ViT-B/16	Vision Transformer with self-attention on 16 × 16 patches.	0.6386	0.5982	6,108
03_5	ResNet50	Deeper residual network; standard 224 × 224 cropping.	0.6707	0.6415	5,566
03_5	ConvNeXt-Tiny	Modern 7 × 7 depthwise conv backbone; standard cropping.	0.6739	0.6458	5,492
03_7	**ConvNeXt-Tiny**	**Smart white-border cropping, layer-wise LR, power-0.25 sampler, flip TTA.**	**0.6998**	**0.6752**	**5,043**

> **From the standard to the optimised ConvNeXt-Tiny:** going from 03_5 (0.6739 Weighted F1) to 03_7 (0.6998) adds **+2.59 points Weighted F1** and **+2.94 points Macro F1**. This gain combines four changes: the smart white-border crop, layer-wise learning rates, the balanced sampler and flip TTA. The notebooks do not measure the crop on its own; flip TTA alone adds about 0.2 points (0.6980 → 0.6998).

5.2 Grad-CAM Visual Interpretability Analysis

To evaluate what the optimized ConvNeXt-Tiny network attends to, we implemented Gradient-weighted Class Activation Mapping (Grad-CAM) targeting the last ConvNeXt stage (`features[7]` in torchvision) [notebook `03_8_image_convnext_gradcam.ipynb`].

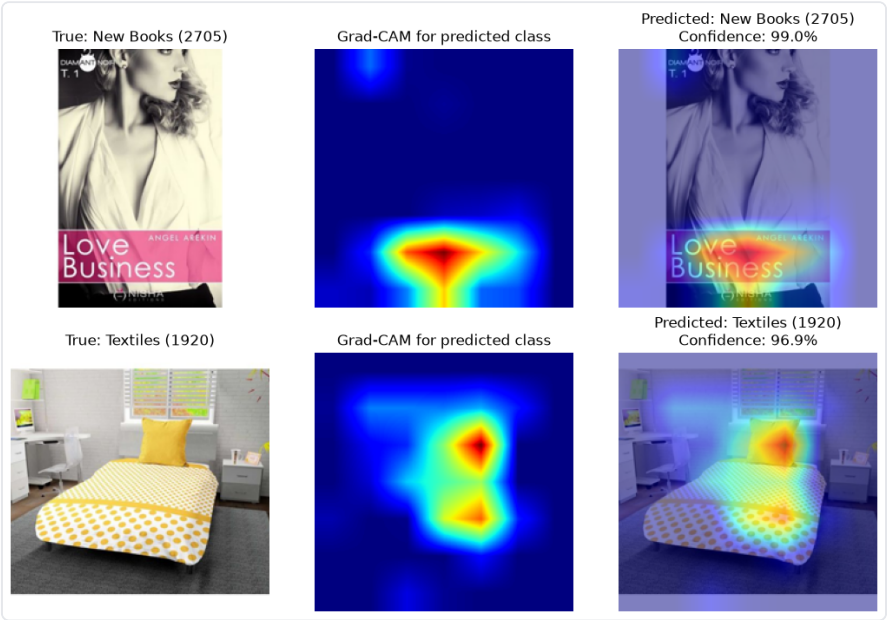


Figure 3: Grad-CAM for two correct predictions: a New Books cover (99.0% confidence) and a Textiles bed scene (96.9%). The heat sits on the cover title and on the pillow and bedding, not on the background.

- **Correct predictions (Figure 3):** for the New Books cover, the heat sits on the title band; for Textiles, on the pillow and the bedding.
- **Effect of smart cropping (Figure 4):** without the crop, the book is small, the heat spreads over the white background and the model predicts Stationery (73.6%). With the crop, the heat moves to the cover title and the model predicts the correct Used Books (97.2%).
- **Errors (Figure 5):** a used book is predicted as New Books (94.4%): the model looks at the cover title, but a cover cannot show whether a book is used. A crayon set (Toys) is predicted as Stationery (98.0%), because crayons fit both categories.

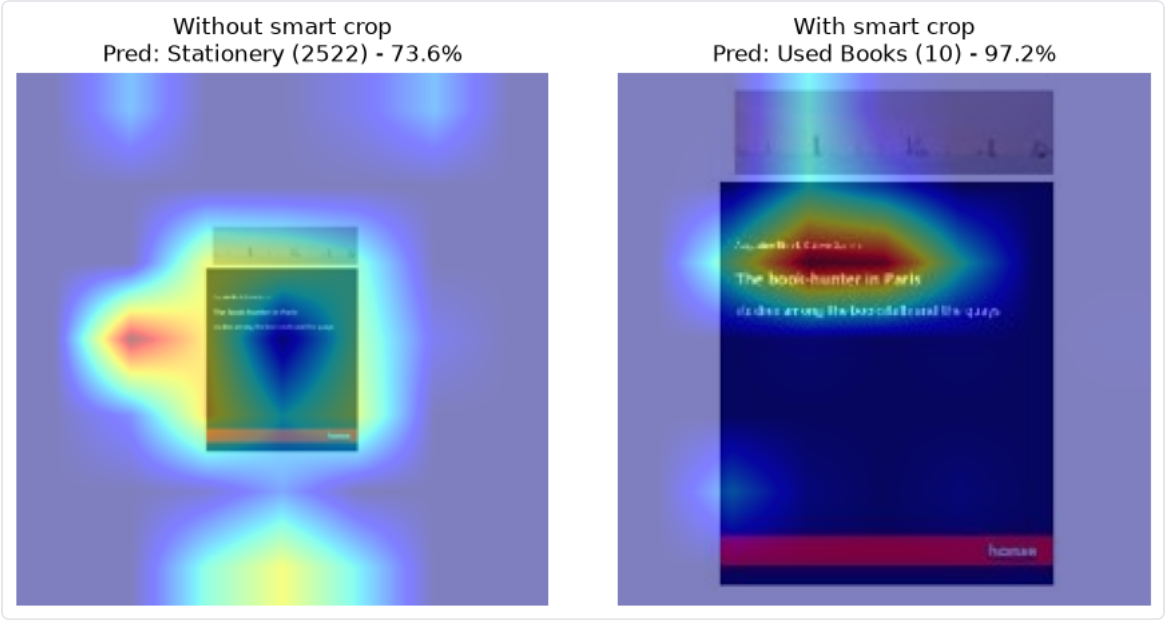


Figure 4: The same validation image without and with smart cropping. Without the crop the heat spreads over the white background; with it, the heat moves to the cover title.



Figure 5: Grad-CAM on two errors: a used book predicted as New Books, and crayons (Toys) predicted as Stationery.

6. Multimodal Fusion & Empirical Results

6.1 Fusion Formulations and Out-of-Fold Evaluation

For every listing in the 16,984 validation set, our unimodal models yield three calibrated class probability distributions: P_{SVM}, $P_{\text{CamemBERT}}$, and P_{ConvNeXt}. We evaluated two late fusion strategies:

1. **Soft Voting:** Linear combination of probabilities:

$$\hat{y} = \text{argmax}_c \sum_m w_m P_m(c)$$

   We evaluated equal weighting (w = [1/3, 1/3, 1/3]) and grid search weighted voting over 5-fold CV.
   Weighted soft voting: each model gets its own weight (steps of 0.1, summing to 1), chosen on the 4/5 part of each fold only. The folds chose 0.4 for CamemBERT, 0.3 for the SVM and 0.3 for the image (0.8920 Weighted F1). Equal weights score slightly higher (0.8941), because the 0.1 grid cannot choose 1/3. The gap is within noise, so tuning the weights adds nothing: the three models are already well balanced.

2. **Meta-Classifier Stacking (Random Forest):** Concatenating the three distributions into an 81-dimensional meta-feature vector:

$$\mathbf{x}_{meta} = [P_{CamemBERT} \,||\, P_{SVM} \,||\, P_{ConvNeXt}] \in \mathbb{R}^{81}$$

   A Random Forest meta-classifier (300 trees, `min_samples_leaf=2`) learns non-linear gating functions to decide which modality to trust conditioned on predicted class confidence.

Model Architecture	Modality	Weighted F1	Macro F1	Δ vs CamemBERT	Errors (/ 16,984)
ConvNeXt-Tiny (03_7)	IMAGE	0.6998	0.6752	–17.67	5,043
Calibrated LinearSVC (03_2)	TEXT	0.8430	0.8326	–3.35	2,656
Weighted Soft Voting (SVM + Image) (04)	FUSION	0.8640	0.8548	–1.25	2,278
CamemBERT-base (03_3)	TEXT	0.8765	0.8594	Baseline	2,082
Weighted Soft Voting (CamemBERT + Image) (04)	FUSION	0.8882	0.8730	+1.17	1,875
Weighted Soft Voting (CamemBERT + SVM + Image) (04)	FUSION	0.8920	0.8797	+1.55	1,813
Soft Voting, Equal Weights (all 3) (04)	FUSION	0.8941	0.8826	+1.76	1,776
Random Forest Stacking (all 3) (04)	FUSION	**0.8988**	**0.8856**	**+2.23**	**1,715**

6.2 Performance Gain on Products With vs. Without Description

A pivotal question for e-commerce cataloging is how multimodal fusion performs when product descriptions are absent. The empirical breakdown across the 16,984 validation listings provides unambiguous evidence:

Model Architecture	Overall F1 (16,984)	With Description (11,048)	No Description (5,936)	Modality Drop (Δ)
ConvNeXt-Tiny (Image)	0.6998	0.6845	0.7351	+0.0506
LinearSVC (Text)	0.8430	0.8768	0.7796	–0.0972
CamemBERT (Text)	0.8765	0.8996	0.8349	–0.0647
Soft Voting Equal Weights	0.8941	0.9050	0.8743	–0.0307
Random Forest Stacking	**0.8988**	**0.9120**	**0.8739**	**–0.0381**

> **Key Finding on Missing Descriptions:** When descriptions are missing (35.0% of products), LinearSVC F1 plunges by **9.72 points** and CamemBERT drops by **6.47 points**. Meanwhile, the image model achieves **0.7351 F1** on these products (book covers and packaging are highly informative). Consequently, Random Forest fusion gains **+3.90 F1 points over CamemBERT alone** (0.8349 → 0.8739) and **+9.43 F1 points over LinearSVC**. The visual stream serves as a resilient safety net for title-only catalog listings.

6.3 Per-Category Synergies and Delta Distribution

Evaluating the Random Forest fusion against CamemBERT across all 27 categories reveals remarkable consistency with Figure 6:

- **25 of 27 categories improve** over CamemBERT alone.
- **1 category stays equal:** PC Games (2905).
- **Exactly 1 category exhibits a minor drop:** Baby Socks (1301, –0.3 F1 points / –0.0033).
- **Games (code 40):** CamemBERT F1 is 0.7953, and Best Fusion reaches 0.8546, representing an exact gain of **+5.9 F1 points (+0.0593)**, matching Figure 6 perfectly.
- **Largest gains:** New Books (2705, **+10.8 points**), Used Books (10, **+9.9**), Wargames (1180, **+9.3**), Games (40, **+5.9**) and Outdoor Toys (1302, **+4.3**). The image tells new books from used ones, and the SVM catches the rare exact words of Wargames.

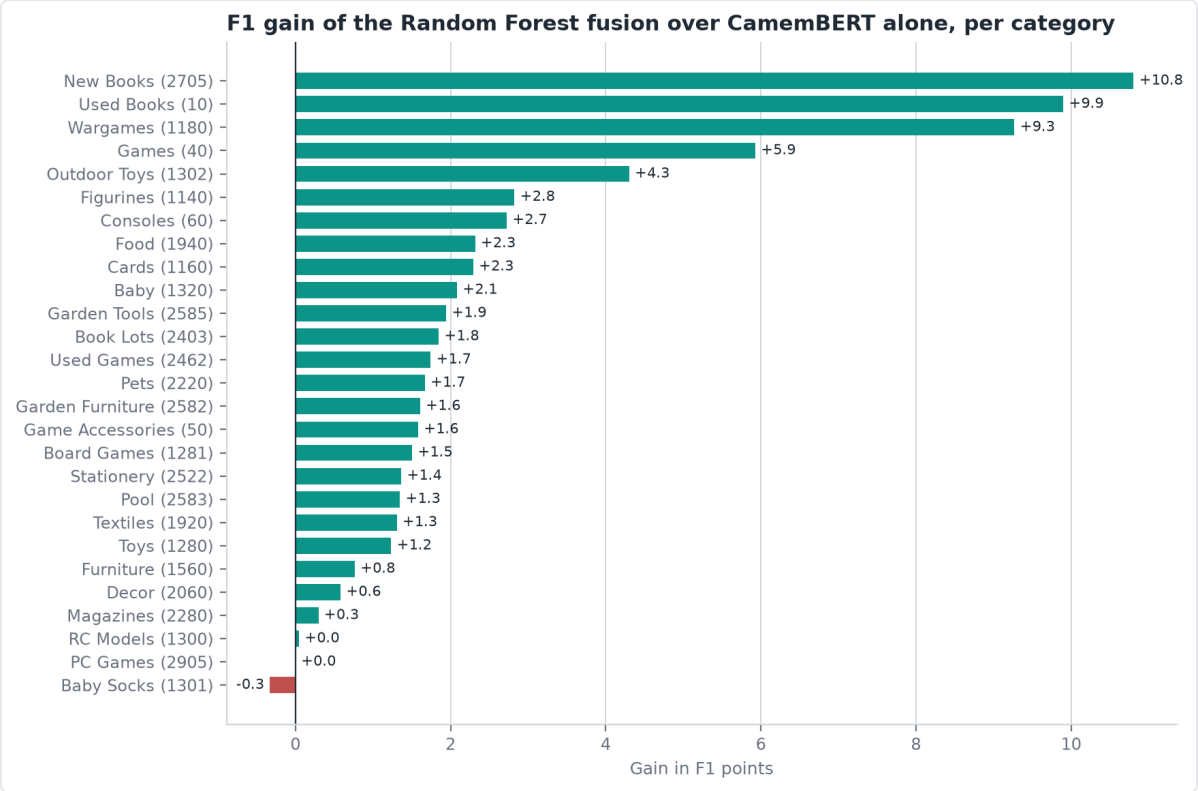


Figure 6: F1 score delta of the Random Forest fusion over CamemBERT alone, per category. 25 categories improve, 1 stays equal, and 1 declines slightly (–0.3).

6.4 Confusion Matrix and Error Patterns

Figure 7 displays the normalized out-of-fold confusion matrix of the Random Forest fusion across all 27 categories:

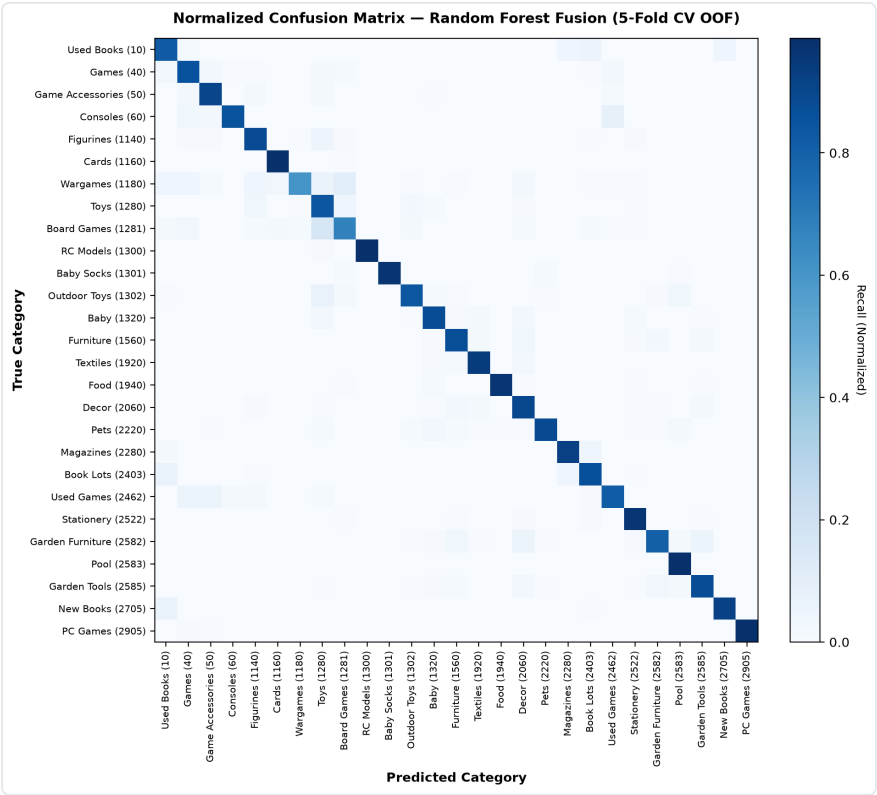


Figure 7: Normalized confusion matrix of the Random Forest multimodal fusion (5-fold cross-validation out-of-fold predictions on the 16,984 validation products).

True Category	Predicted Category	Errors	Domain Nature of Misclassification
Board Games (1281)	Toys (1280)	61	Cardboard packaging overlap; toy brands marketing licensed board games.
Toys (1280)	Board Games (1281)	56	Reciprocal packaging confusion between toy kits and family games.
Book Lots (2403)	Used Books (10)	52	Volume bundles vs. single vintage novels; identical cover artwork.
Book Lots (2403)	Magazines (2280)	41	Serialized paperbacks sharing periodical format and binding.
Furniture (1560)	Decor (2060)	41	Small shelving, bedside lamps, and decorative storage.
Toys (1280)	Figurines (1140)	39	Franchise action figures categorized ambiguously as toys vs. collectibles.
New Books (2705)	Used Books (10)	38	Identical title strings; listing fails to specify condition in text.
Magazines (2280)	Book Lots (2403)	38	Multi-issue magazine bundles classified as book lots.

The confusion matrix confirms that remaining misclassifications are almost exclusively confined to **semantically adjacent sibling classes** within the same broad domain (e.g., book bundles vs. individual used books; board games vs. children's toys). Cross-domain errors (e.g., confusing garden pools with video games) have been virtually eliminated by the multimodal fusion.

7. Feature Attribution & Model Interpretability

7.1 Text Feature Attribution: Distinctive vs. Shared Vocabulary

Inspecting LinearSVC feature weights [notebook `05_interpretability.ipynb`] reveals why classical linear models struggle on book sub-genres:

- **Category-Defining Vocabulary:** Unambiguous categories possess high-weight, exclusive n-grams: *piscine* and *intex* for Pool (2583), *chien* and *laisse* for Pets (2220), *warhammer* and *heroclix* for Wargames (1180).
- **Shared Book Vocabulary:** Generic French literary tokens (e.g., *livre*, *roman*, *tome*, *edition*) carry nearly identical positive weights across Used Books (10), Magazines (2280), Book Lots (2403), and New Books (2705). In the absence of a publisher date or lot descriptor, linear text models cannot resolve condition or format.

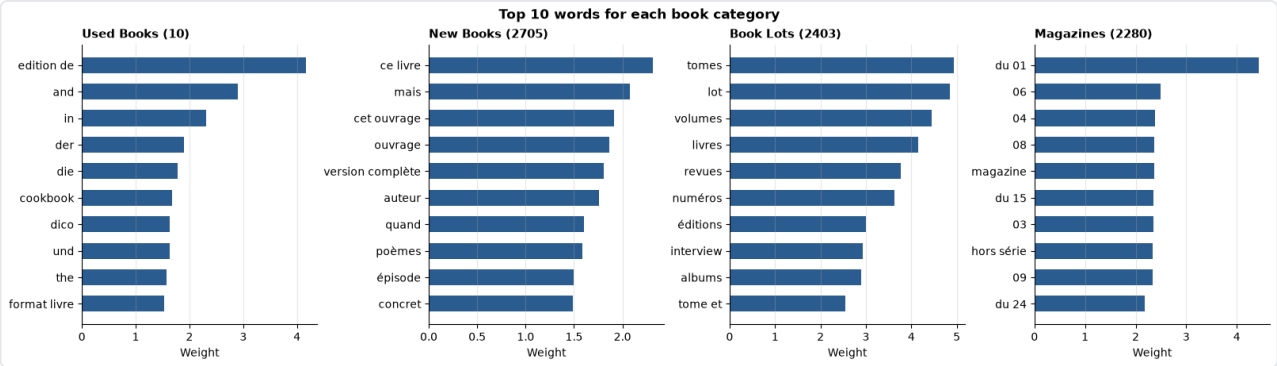


Figure 8: Top LinearSVC feature weights across the four book categories. Strong discriminative tokens are scarce; generic literary vocabulary is shared.

CamemBERT word importance: CamemBERT has no word weights, so we remove one word at a time and measure how much its confidence drops (Figure 8b):

- **Book Lots, CamemBERT right:** for *les chemins de maison haute 1 & 2*, removing *1* lowers its confidence by 18 points. CamemBERT reads *1 & 2* as two volumes, a lot of several books. The SVM predicted Used Books.
- **Wargames, CamemBERT wrong:** for *chaos chosen age of sigmar*, the word *of* matters most (33 points). CamemBERT reads *age of …* like a video game title and predicts Games. The SVM gets it right through *sigmar*, a rare Warhammer word.

- The two text models fail on different products, which is why combining them helps.

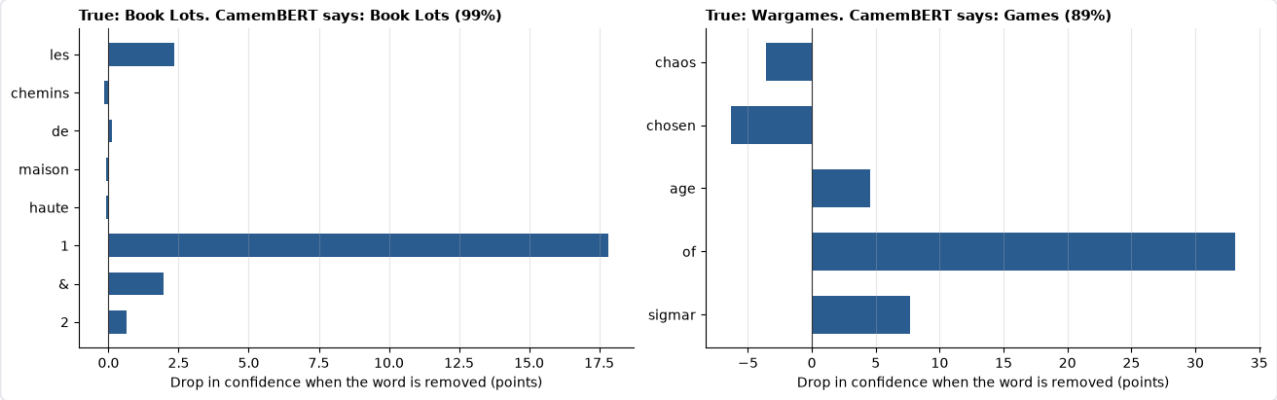


Figure 8b: Drop in CamemBERT's confidence when each word is removed, for one title it classifies correctly (left) and one it gets wrong (right).

7.2 Random Forest Meta-Feature Attribution

Analyzing Gini feature importances within the Random Forest stacking meta-model (Figure 9) confirms how multimodal responsibilities are allocated:

- **CamemBERT:** Accounts for **66%** of the feature importance, acting as the primary classifier.
- **LinearSVC:** Accounts for **23%** of feature importance, capturing exact rare n-grams and brand identifiers that subword embeddings occasionally dilute.
- **ConvNeXt-Tiny:** Accounts for **11%** of the importance. Removing it still costs 1.2 points, almost all on New Books (+11.4) and Used Books (+10.1) (Figure 10).

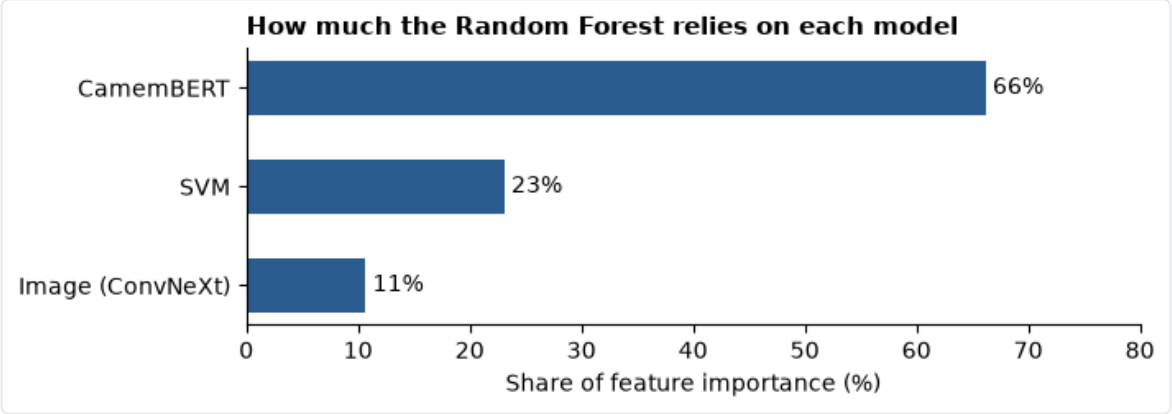


Figure 9: Relative share of Random Forest feature importance contributed by each model's probability outputs.

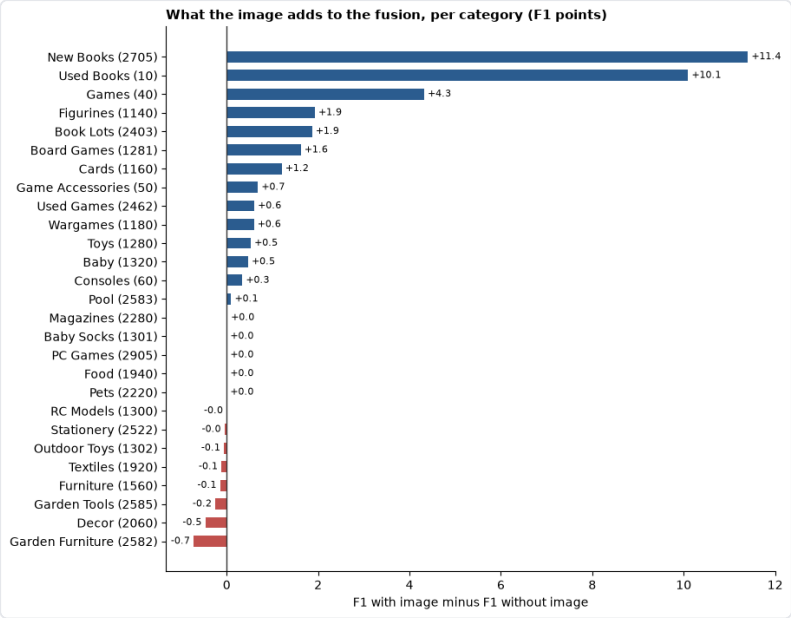


Figure 10: Category-level F1 delta of the Random Forest with the image stream versus without the image stream (5-fold cross-validation).

8. Production Deployment & Recommendations

8.1 Production Architecture & Engineering Trade-offs

Deploying multimodal deep learning models into production requires balancing accuracy against inference latency, financial compute costs, and infrastructure complexity:

Deployment Tier	Target Architecture	Weighted F1	Compute	Recommended Use Case
Tier 1: Ultra-Fast	Calibrated LinearSVC (Text only)	0.8430	CPU only	Real-time seller form autocomplete; bulk backfill of legacy text catalogs.
Tier 2: Intermediate	Equal Soft Voting (all 3)	0.8941	GPU for CamemBERT and ConvNeXt	Zero meta-model maintenance; high accuracy without training a secondary stacking layer.
Tier 3: Maximum Accuracy	**Random Forest Stacking (all 3)**	**0.8988**	**GPU for CamemBERT and ConvNeXt, plus the Random Forest**	**Production marketplace pipeline; optimal resolution of ambiguous listings.**

Recommended Production Architecture — Cascaded Inference: Rather than evaluating heavy neural pipelines on all listings, production systems should adopt a two-stage cascade:

1. Pass new listings through the calibrated LinearSVC. If the top class probability exceeds **0.92** (21% of the validation products), accept its prediction on CPU.
2. For remaining uncertain or title-only listings, invoke the CamemBERT and ConvNeXt-Tiny pipelines and route probabilities through the Random Forest. On the validation products this keeps the full Random Forest score (0.8987 against 0.8988) and runs the GPU models on 79% of products, saving about 21% of GPU work.

8.2 Project Conclusion

This project demonstrates that while text is the dominant modality for e-commerce categorization, computer vision provides a vital, non-redundant signal that compensates for real-world catalog imperfections. By pairing rigorous data hygiene (locked

stratified split, deduplication, smart white-border cropping) with modern deep learning backbones (CamemBERT, ConvNeXt-Tiny) and out-of-fold late fusion, we advanced classification performance from a naive baseline of **0.0258 F1** to a robust **0.8988 Weighted F1** (0.8856 Macro F1). The full pipeline is open, reproducible, and readily adaptable for production deployment.